

Hybrid method to solve a facility layout problem: Genetic Algorithm - Particle Swarm Optimization [★]

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Abstract: In this paper, we solve the Facility Layout Problem (FLP) by using a hybrid method that combines a genetic algorithm (GA) and a particle swarm optimization (PSO). The GA is used to solve the Group Technology Problem (GTP) which groups simultaneously machines and products in cells since the particle swarm algorithm optimizes quadratic assignment problem taking into account the GTP solution. This hybrid method is compared with a previous one, called YMAY2. The obtained results are promising.

Keywords: group technology, facility layout problem, quadratic assignment problem, particle swarm optimization, genetic algorithm.

1. INTRODUCTION

With this paper, we aim to present a method to solve a two steps layout Problem. This approach consists to combine a QAP with a group technology problem called T-FLP as done by [16] [17] [18]. The approach allows to deal efficiently with real life problems specially for the storage area optimization and the setup time consideration for example. A new method is considered in this paper which combine a genetic algorithm and a particle swarm optimization and called (GA-PSO). This one is compared with the previous developed method called YMAY2 (combines a genetic algorithm with an hybrid ant colony optimization method). A large scale of test were applied and the results are very interesting.

The proposed method process in two steps as shown in Figure 1. The first one creates cells with families products and group of machines using input data volume. The second step assigns the machines to positions taking into account all machines belonging to the same cell. A main loop is considered to obtain, over different cells configuration, the best FLP solution.

As known, the facility layout problem (FLP) occurs when we look to reorganize an existing structure or to construct a new one. Recently Drira et al. [1] have made a full description of the problem and resolution approach. Their paper describes workshop characteristics impacting the layout, resolution approach, and different formulation of layout problem. FLP is known to be NP-Hard [25], and in

many times formulated as a quadratic assignment problem (QAP). In [5], they have described all problems that are formulated as QAP.

For solving the FLP, several approaches are used: exact method and approximated ones. For the exact method, the latest work has been presented by [13] which obtain an optimal solution for the two dimensional FLP using a branch and bound algorithm. [24] presented also a branch-and-bound algorithm for a continuous FLP.

For the approximated method, we can quote the work of Liang and Chao in 2008 [11] who have developed a tabu search technique to solve the FLP. Ant colony optimization (ACO) has also been considered for solving FLP by [15], [14]. Hani et al. [25] used an ant colony optimization mixed with guided local search (ACOGLS) for solving an industrial layout problem. Mahdi [9], has resolved the FLP using a simulated annealing mixed with an exact method. The Particle Swarm Optimization (PSO) is not used a lot to solve the FLP, however Christu Paul et al. [19] have implemented a PSO under constraints of passages and structure walls.

For the Group Technology Problem (GTP), it was first proposed by Mc Auley in 1972 [3]. The main objective is to create products families and assign them to groups of machines. For this type of problem, a binary matrix values machine-products is usually used as input data in order to identify machines used to produce product. Various constraints exist as size limits of the cells or cell numbers.

There are several resolution approaches for GTP, as those based on similarity coefficient [4]. The analytical approach

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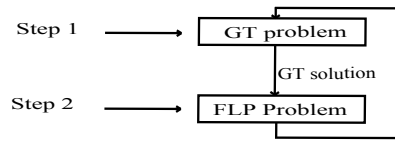


Fig. 1. T-FLP

is most commonly used for grouping machines into cells. [8] summarized the different studies of this type. The metaheuristics are also widely used for GTP, as Genetic Algorithm [7] [6].

The main idea of our approach developed in this paper is how to consider the operational management constraint in the design step of a plant? For that we use a technology group in a facility layout problem [16]. In the literature there exist many methods to solve the QAP and the GTP. For the GTP, it was noticed that binary matrix are generally used as input data to identify similar machines and products. To be as close as possible of the industrial constraints, we purpose to solve the GTP with a data volume production. This had never been used following our resolution approach. For solving the facility layout problem, the QAP is used and takes into account the GTP solution as an initial solution. This new approach was called T-FLP [16] [17] [18]. So to solve it, we have developed and implemented a method called YMAY2 which combines GA to solve the GTP and ACOGLS to solve the QAP. In this paper, we aim to improve the solution obtained by YMAY2 using the PSO to replace the ACOGLS.

This paper is organized as follows: in section 2, the problem formulation is presented. Section 3 gives a brief description of the GA and the PSO used. Section 4 is dedicated to the results and section 5 presents the conclusion.

2. PROBLEM DESCRIPTION AND FORMULATION

The aim of our study is to operate on the strategic and tactical plan to reorganize a workshop. It contains three evaluations. The first evaluates the quality of the cells created in the first step. In the second step, we evaluate the quality of the assignment of the machines in the locations with taking into account those belonging in the same cell.

This section will define the mathematical models used to solve our problem. At first the mathematical model of the group technology problem will be described. The second subsection presents the quadratic assignment problem.

2.1 Group technology approach

The goal of solving a group technology problem is to minimize inter-cell movements between cells, a binary matrix is usually used as input data. In our problem, a volume matrix has been used. The mathematical model used to solve our problem was inspired by Chan *et al.* (1998). The Objective function has been changed to take into account

the inter-cell volume transported and maximum and minimum number of machines per cells has been integrated as constraints. [18]

The objective function value is:

$$\text{Max } Z_{MPGV} = C_{int}Q_{int} - (1 - C_{int})Q_{ext} \quad (1)$$

Where $C_{int} \in [0, 1]$ is the coefficient of importance of internal volume of production which is generally considered between $\{0.5, \dots, 0.7\}$ [7] [6].

The production volume inside the cells Q_{int} :

$$Q_{int} = \sum_{k=1}^L \sum_{j=1}^P \sum_{i=1}^M Q_{ij}x_{ik}y_{jk} \quad (2)$$

Where :

i, j, k : is the indices respectively of machines, products and cells.

M, P, L : is the respectively the maximum number of machines, products, and cells.

x_{ik} and y_{jk} is equal to "1" if machine i respectively product j is assigned to cell k , and "0" else.

Q_{ext} calculates the inter-cell volume transported and described as follow:

$$Q_{ext} = \frac{\sum_{i=1}^L \sum_{j=1}^P Q_{ij} \sum_{k=1}^M |x_{ik} - y_{jk}|}{2} \quad (3)$$

Q_{ij} represents the flow matrix indicating the volume between machine and products.

As constraints :

$\sum_{k=1}^L x_{ik} = 1 \forall i$	each machine must be assigned only in one cell
$\sum_{k=1}^L y_{jk} = 1 \forall j$	each product must be assigned only in one cell
$\sum_{k=1}^L \sum_{i=1}^M x_{ik} < (\beta_1 M)$	each cell must not exceed $\beta_1 M$ machine
$\sum_{k=1}^L \sum_{i=1}^M x_{ik} > \beta_2$	each cell must contain at least β_2 machines to ensure the cell density.
$\beta_1 \in [0, 1]$ and $\beta_2 \in [1, M]$	which is proportional to the cell number(to create).

The following section presents the second step of our method which is a solution of assignment machines in locations solved by a quadratic assignment problem.

2.2 Quadratic assignment problem

The QAP is first introduced by Koopmans and Beckmann in 1957 and used traditionally to solve the facility layout problem. Liolia *et al.* [5] described problems formulations. QAP is a problem of allocating a set of machines to a set of locations in order to minimize the total cost associated

to distances between locations and flows.

The objective function value is defined by [5]:

$$\text{Min } Z_{QAP} = \sum_{i,j=1}^n \sum_{k,p=1}^n f_{ij} d_{kp} x_{ik} y_{jp} \quad (4)$$

with :

$$\sum_{i=1}^n x_{ij} = 1, \sum_{j=1}^n x_{ij} = 1, x_{ij} \in \{0, 1\} \quad 1 \leq i, j \leq n$$

f_{ij} represents the flow between facilities i and j .
 d_{kp} represents the distance between locations k and p .
 x_{ij} defines the availability or not of resources in the location.

3. RESOLUTION METHOD

This section contains a presentation of the implemented methods to solve our problem. At first a genetic algorithm used to solve the GT problem is presented, and after the Particle swarm optimization is described. We have chosen a genetic algorithm since it has given very interesting results for solving a group technology problem [6] [18]. The particle swarm optimization is used to solve the QAP since it has given a very efficient results for the standard FLP [12], [23] and to test its efficiency on this type of decomposition method.

3.1 Genetic algorithm

The GA used for solving the problem is inspired by the works of Chan et al. (1998,2006). Several changes have been operated compared with the original algorithm. The efficiency of the changes was presented in [18].

This section presents parameters of the GA and the change operated on the original algorithm. Each chromosome in the population represents a solution for the GTP. Each chromosome is composed by a part of machines and a part of product. The size (number of gene) of chromosome is the sum of the number of machines and the products considered. Each gene of the chromosome contains an integer number that represents the indices of cell in which machines or products are assigned. The initial solution is obtained by randomly assigning machines to cells. After that, the products are sorted in volume decreasing order and are finally assigned to the cells following their machine ranges (Figure 2).

Fig. 2. Chromosome encoding

machine/product	M1	M2	M3	P1	P2	P3	P4
Cell number	2	1	3	2	1	1	3

The reproduction uses a crossing process which consists in obtaining two children by crossing one or more genes from parents. For solving our problem, data oriented approach has been used. A cross at one point is made on the machines, and the products are assigned according to their job sequences.

The following algorithm resume the algorithm where the changes operated are in bold.

Algorithm 1 GA algorithm [18]

Step 1: Generate random initial population of n chromosomes.

Step 1.1: Assign randomly machine to cells.

Step 1.2: Sorting the product according to their volumes decreasing.

Step 1.3: Assign the product to the cells following their machines ranges.

for each iteration do

Step 2: Calculate the objective function Z_{MPGV} for each chromosome.

Step 3: Check if all the solutions of the population are achievable, else Penalize them:

Step 3.1: Check that each machine contains at least one product in his cell, which uses it.

Step 3.2: Check all products that contain in their cells at least one machine in his cell, part of their ranges.

Step 4: Penalize the infeasible solutions

Step 5: Create the new population:

Step 5.1: Random selection of the best chromosomes from the value of the fitness function

Step 5.2: Apply a cross using a data oriented approach on the chromosomes selected with a probability.

Step 5.3: Apply a mutation on chromosomes selected with a probability.

Use the new chromosomes in the new generation.

Print the best solution if the criterion for judgment is reached, else go to *step 2*.

end for

3.2 Particle swarm optimization

Particle Swarm Optimization (PSO) is a stochastic optimization technique and also a population based search algorithm first proposed by Eberhart and Kennedy (1995) [10], [21]. It is inspired from the bee population behavior.

[While humming, she describes a kind of eight flattened slightly tilted. Undulating quickly, it runs once, twice, ten times, the same closed curve, followed by her sisters who are close to her,... and listen: the bee dances. And his dance is a message that tells them: "15° to the sun / 300 meters / lavender / a lot"]. This small paragraph is extracted from [12] and translated from french. It means that bees transmit information between them, and this is the base of the algorithm. The efficiency of the algorithm is based on the collaboration and not the competition too.

Each particle has got a velocity, a memory and informants. The positions and the velocity of the first population of particles is generated randomly. To decide to move, the particle needs four informations: her actual velocity, her best performance and the best performance of her neighbors. It is based on three parameters w , $c1$ and $c2$ which allow them respectively either to follow its own path, retrace his steps, or follow the best neighbor.

For our problem, each solution contains the positions of the machines in the workshop as described in Figure 3.

Fig. 3. Particle encoding

Machine	M1	M2	M3	M4	M5	M6	M7
Position	2	1	3	4	7	6	5

The Algorithm 2 resumes the PSO algorithm.

Algorithm 2 PSO algorithm [19]

- 1: Step 1: Generate initial solution randomly for all particles.
- 2: Step 2: Assign $Pbest[i] = \text{initial solution where } i = 1, 2, \dots, N$ (N : nb of particles).
- 3: Step 3: Find best among all particles and assign this to $Gbest$.
- 4: Step 4: Generate initial velocities randomly for all particles.
- 5: Step 5: Add velocities to the corresponding particles, i.e.,
 $Present[i] \text{ (new)} = Present[i] \text{ (old)} + V[i]$.
- 6: Step 6: Update velocity according to
 $V[i] = w * V[i] \text{ (present)} + c1 * (Pbest[i] - present[i]) + c2 * (Gbest[i] - present[i])$.
- 7: Step 7: if number of iterations < cyc Goto step5.
- 8: Step 8: Ubest is the best among all $Gbest$ If number of iterations j cond Goto step1.
- 9: Step 9: Write Ubest. Stop.

Like described before and following the algorithm 2, the particles of a swarm decide to move by using the information of the:

- Velocity of the particle i in iteration k V_i^k .
- The best position of the particle i (local best) ($Pbest_i$).
- The best position of the particle group (global best) ($gbest$).
- And the position of the i^{th} particle in k^{th} iteration x_i^k ($k = 0, \dots, itermax$ and $i = 1, \dots, N$).

4. COMPUTATIONAL EXPERIMENTATION

The method is implemented using Visual C++ 2005 on an Intel Core Duo with 1.6 Ghz CPU speed. All parameters of the GA was tested in [17] and fixed, so the Table 1 resumes them. The algorithm is evaluated about 5 runs to test the different configurations because the particularity of the GT algorithm that is done the same solution with different configurations. Those configurations have an impact on the solution of the FLP.

Table 1 presents the different parameters of the GA.

Table 1. Parameters of GA

Parameters	Value
<i>Number of iterations</i>	500
<i>Populationsize</i>	50
<i>C_{int}</i>	0.7
<i>P_{croi}</i>	0.8
<i>P_{mut}</i>	0.2

The PSO is tested on a population between 10 and 500 and fixed to 200. The values $w, c1, c2$ are proposed by (Clerc et al. 2003) with $w = 0.9$ and $c1 = c2 = 1.7$. max_{iter} which represents the maximum number of iterations is fixed as 80000.

Table 2 presents the different parameters of the PSO.

Table 2. Parameters of PSO

Parameters	Value
N	200
w	0.9
$c1, c2$	1.7
max_{iter}	80000

The results obtained by the methods are shown in the Table 3. $Problem_i$ is a problem randomly generated where i defines the size of the problem (i.e. the number of machines). The different matrix are generated using a uniform distribution flow (between 0 and 10), volume (between 0 and 100) and distance (between 2 and 40).

Z_1 is defined in Eq.(1) which maximizes the volume belonging the cells and Z_2 is defined in Eq.(4) which minimizes the distance between the machines.

Each problem ($Problem_i$) contains 10 instances to ensure the quality of the solution. The two methods were tested on **120** instances.

To compare the two methods, we have three gaps:

$$GAP_1 = \frac{Z_{QAP}(Y MAY2) - Z_{QAP}(GA-PSO)}{Z_{QAP}(Y MAY2)}$$

$$GAP_2 = \frac{Z_{QAP}(proctime[GA-PSO]) - Z_{QAP}(proctime[Y MAY2])}{Z_{QAP}(proctime[GA-PSO])}$$

$$GAP_3 = \frac{Z_{QAP}(proctime[PSO]) - Z_{QAP}(proctime[ACOGLS])}{Z_{QAP}(proctime[PSO])}$$

GAP_1 : shows the gap between the objectif function of the GA-PSO method and YMAY2.

GAP_2 : shows the gap between the processing time of GA-PSO and YMAY2.

GAP_3 : shows the gap between the processing time of the PSO and the ACOGLS.

Table 3. Results

	GAP(%)		
	GAP_1	GAP_2	GAP_3
Problem 2	0	77	78
Problem 3	0	58	57
Problem 4	0	72	73
Problem 5	0	86	86
Problem 6	-0.33	80	80
Problem 7	-0.47	70	71
Problem 8	-1.40	64	64
Problem 9	-1.09	38	36
Problem 10	-2.49	41	40
Problem 11	-0.41	29	26
Problem 12	-0.69	13	9
Problem 13	-0.04	-6	-13

For the problems between problem 2 and 5, the solutions of the two algorithm have the same performance. For the other problems range, we notice that the GA-PSO solutions outperform those obtained by YMAY2 method with an average gap of 1%, with 2.49% for some cases.

For the computational side, YMAY2 is mainly faster with a gap performance around 50% but GA-PSO is better in some cases with a gap of 6%. We notice that there is relationship between the algorithm speed and the problem size in sense that more the problem is bigger more the GA-PSO is performer.

5. CONCLUSION

We deal in this paper with the workshop design approach called T-FLP. The main idea is to place machines in work-

shop with taking into account technology information. The new approach combines between a Group Technology Problem (GTP) and a Facility Layout Problem (FLP). The goal of this work is to compare the GA-PSO method to our previous method YMAY2. The GT solution proposed to group similar entities according to their job sequences, and their volumes exchanged, which reduce the storage area. The GA-PSO has a performance around 1% for the cases considered without any hybridization. This allows us to improve the quality by combining the PSO with a local search procedure.

In perspectives of this work, it would be interesting to treat the multiobjective approach.

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